



BIZRA FARMS INTEGRATED NIGERIA LIMITED

(RC 1929412)

EconomicBridge — Executive Brief

A live satellite and AI platform that turns Earth observation into decisions for agriculture, food security and disaster response. Built and operated in Nigeria by Bizra Farms Integrated Nigeria Limited.

142	18.3M	64%	8
LGAs read pixel by pixel	hectares of farmland greening, 2026 rains	change precision, random sample	live data feeds

Farmland: cropland and rangeland on Esri's 2023 land-cover map; tree canopy and built-up land excluded. Precision: of the land changes the platform reports, the share confirmed by eye against before-and-after Sentinel-2 imagery on a random sample (95% interval 45–80%). Validation against the ground is the next step.

What it does

Farmland Protection — every pixel of every pilot LGA read each rainy season, and each land change on cropland or rangeland reported at its own coordinates, for early warning of encroachment and farmer-herder conflict.

ShockGuard — storms and extreme rainfall rebuilt from half-hourly NASA satellite rainfall. **CropGuard** — crop-disease diagnosis from a leaf photo, validated on laboratory images, plus satellite vegetation monitoring.

Economic Visibility — poverty and population mapping from space. Plus mobility and education intelligence. Rainfall advisories reach farmer cooperative leaders in Kebbi by SMS in Hausa, no smartphone needed.

Built on

Open data from Copernicus Sentinel-1 and Sentinel-2, NASA, Esri land cover, the World Bank, UNICEF and WorldPop; a trained ResNet-50 model plus land-change and anomaly detection; running on AWS with multi-tenant data isolation and honest measured-versus-modelled labelling.

Proposed collaboration with NASRDA

- A **single-state data pilot**: NigeriaSat / NCRS imagery for one cloud-limited state (Benue is the clearest case), measured against the same random-sample standard.
- **Ground-truth validation**: NASRDA field data to validate the land-change detector and the crop-disease model in the field.
- A live **reference platform** for NASRDA's application centres.
- A path to a **data-sharing memorandum of understanding**.